



EKSAMENSOPPGÅVE/EKSAMENSOPPGAVE

Emnekode: MAS 149

Emnenamn/Emnenavn: STATIKK OG FASTHETSLÆRE

Utdanning/kull/klasse: Diverse

Dato: 08.06.2026

Eksamensform: Skriftlig

Eksamenstid: 2 timer

Tal på eksamensoppgåver/Antall eksamensoppgaver: 3

Tal på sider (medrekna denne)/Antall sider (inkludert denne): 12

Tal på vedlegg/Antall vedlegg: 5

Tillatne hjelpemiddel/Tillatte hjelpemidler: Enkel kalkulator

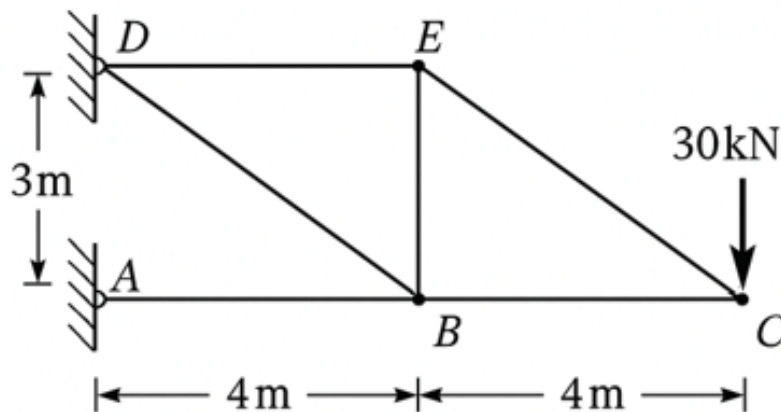
(ikke programmerbar)

Fagansvarleg/Fagansvarlig: Javier de Elio Medina (41259117)

English version

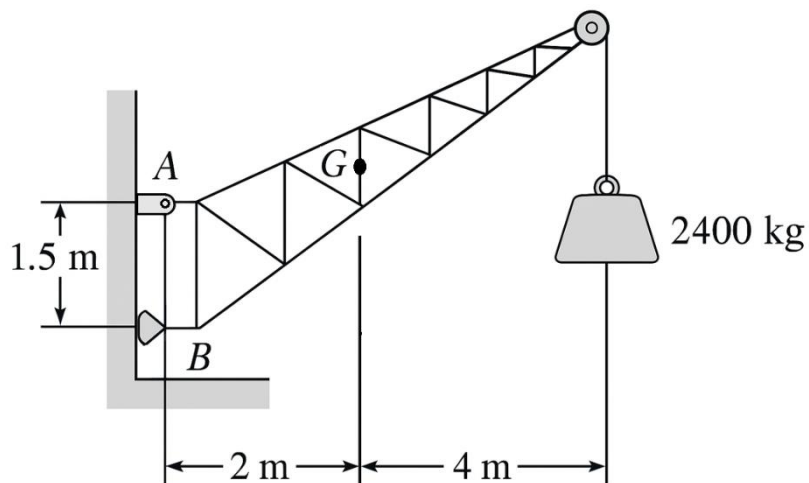
Exercise 1 (35%):

Determine the force in each member of the truss. State if the members are in tension or compression.



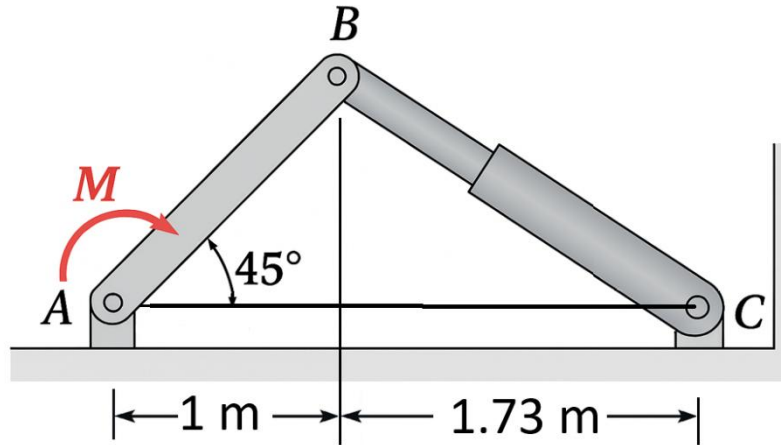
Exercise 2 (35%)

A fixed crane has a mass of 1000 kg and is used to lift a 2400 kg crate. It is held in place by a pin at A and a rocker at B. The centre of gravity of the crane is located at G. Determine the components of the reactions at A and B.



Exercise 3 (30%)

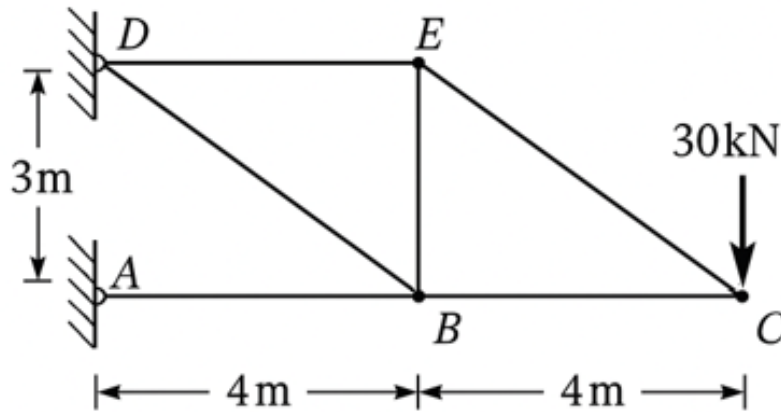
The hydraulic cylinder BC exerts a force P on member AB to maintain equilibrium, as shown in the figure. Given that the magnitude of the moment M is 1200 [Nm], determine the magnitude of the force P .



Bokmål versjon

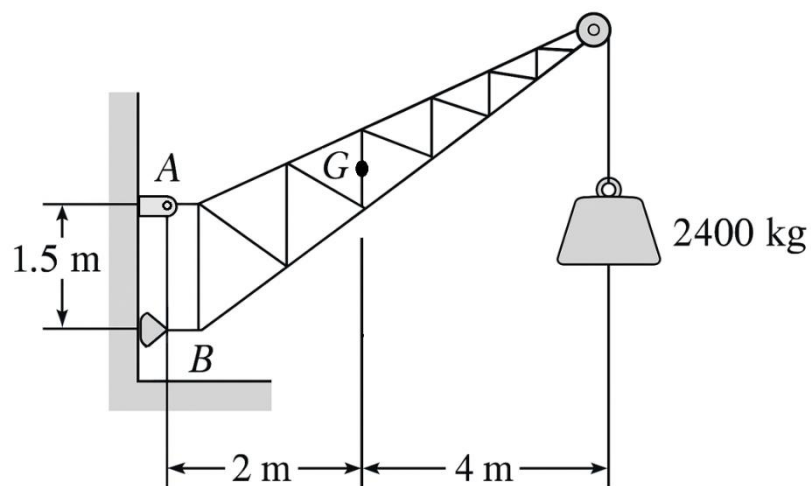
Oppgave 1 (35%):

Bestem kreftene i hver stav i fagverket, og angi om stavene er i strekk eller i trykk.



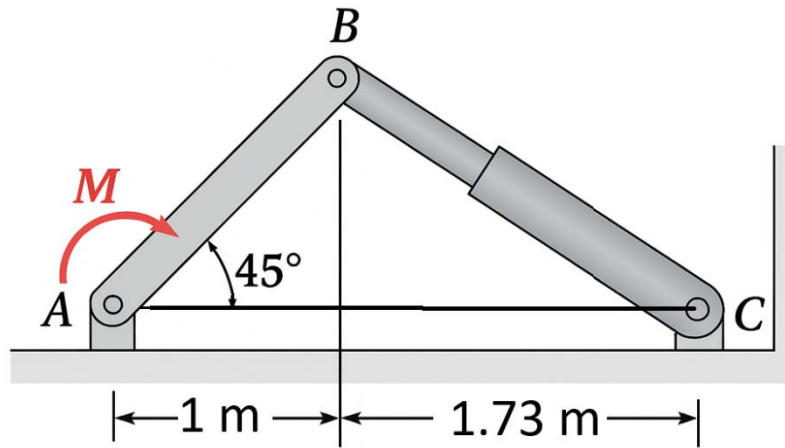
Oppgave 2 (35%)

En fastmontert kran har en masse på 1000 kg og brukes til å løfte en kasse med masse 2400 kg. Kranen holdes på plass av et boltelager i A og et rullelager i B. Tyngdepunktet til kranen er plassert i G. Bestem komponentene av reaksjonskreftene i A og B.



Oppgave 3 (30%)

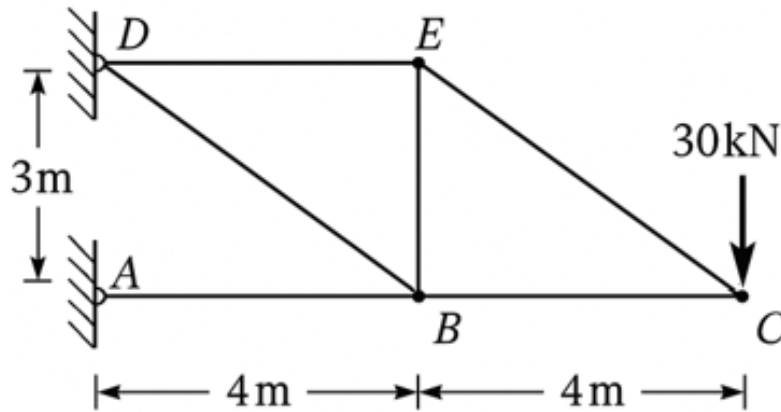
Den hydrauliske sylindren BC utøver en kraft P på staven AB for å opprettholde likevekt, som vist i figuren. Gitt at momentet M har størrelsen 1200 [Nm] , bestem størrelsen av kraften P .



Nynorsk versjon

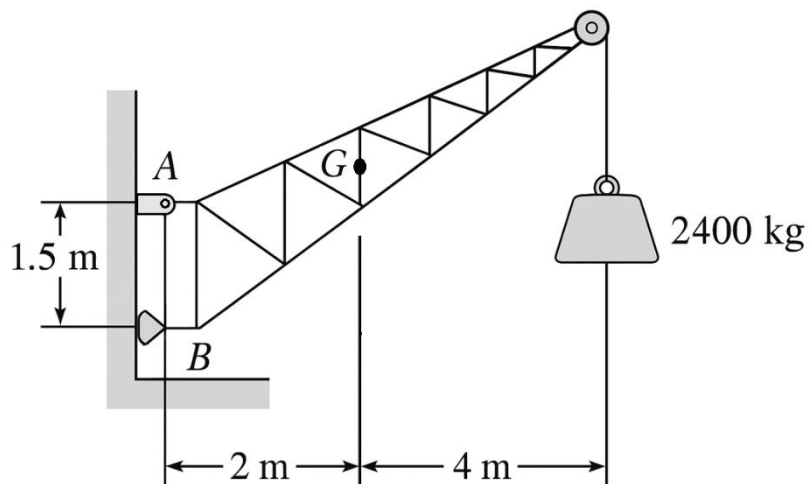
Oppgåve 1 (35%):

Bestem kreftene i kvar stav i fagverket, og oppgi om stavane er i strekk eller i trykk.



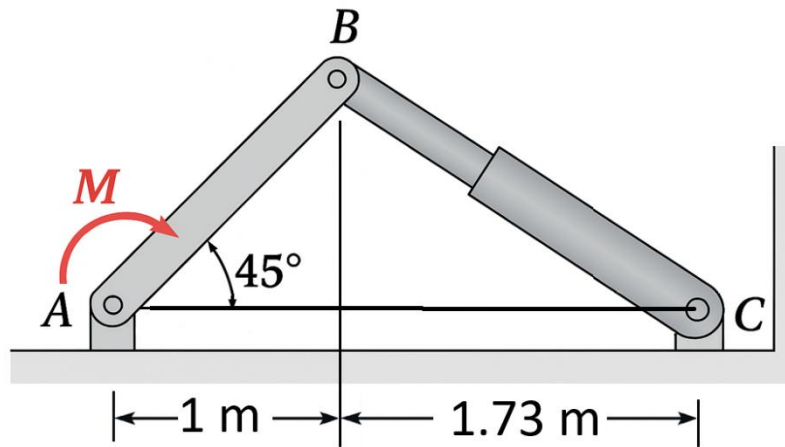
Oppgåve 2 (35%)

Ei fastmontert kran har ein masse på 1000 kg og blir brukt til å løfta ei kasse med masse 2400 kg. Krana blir halden på plass av eit boltelager i A og eit rullelager i B. Tyngdepunktet til krana er plassert i G. Bestem komponentane av reaksjonskreftene i A og B.



Oppgave 3 (30%)

Den hydrauliske sylinderen BC utøver ei kraft P på staven AB for å halda oppe likevekt, som vist i figuren. Gitt at momentet M har storleiken 1200 [Nm], bestem storleiken av krafta P .



Eksamen i MAS149 Statikk og fasthetslære, vedlegg

Friksjon:

$$F_{max} = \mu_s \cdot P$$

Varigions prinsipp

$$M_O = \Sigma F_x \cdot y - \Sigma F_y \cdot x = F_R \cdot a$$

Resultantkraft

$$F_R = \sqrt{\Sigma F_x^2 + \Sigma F_y^2}$$

Aksial spenning:

$$\sigma_{avg.} = \frac{P}{A}$$

Elastisk forlengelse:

$$\delta_F = \frac{P \cdot L}{A \cdot E}$$

$$\delta_T = \alpha \cdot \Delta T \cdot L$$

Torsjon:

$$\tau = \frac{T \cdot c}{J}$$

sirkulære tverrsnitt:

$$\text{massivt: } J = \frac{\pi}{2} c^4$$

$$\text{rør: } J = \frac{\pi}{2} (c_y^4 - c_i^4)$$

Vridningsvinkel, ϕ

$$\phi = \int_0^L \frac{T(x)dx}{J(x)G}$$

$$\phi = \Sigma \frac{TL}{JG}$$

Bøying:

$$\sigma_b = \frac{M \cdot c}{I}$$

Sammenheng mellom w, V og M:

$$\frac{dV}{dx} = -w \quad \frac{dM}{dx} = V$$

Symbol	Forklaring	Benevning
F_{max}	maksimal friksjonskraft	[N]
F_R	resultantkraft	[N]
y, x, a	momentarmen til henholdsvis F_y , F_x og F_R	[m]
μ_s	statisk friksjonskoeffisient	
P	normalkraft	[N]
σ_{avg}	gjennomsnittlig normalspenning	[Pa]
σ_b	bøyespenning	[Pa]
σ_s	totale strekkspenninger (bøye og normalspenninger)	[Pa]
σ_t	totale trykkspenninger (bøye og normalspenninger)	[Pa]
d	diameter	[m]
A	tverrsnittsareal	[m ²]
E	elastisitetsmodul	[Pa]
δ_F	defleksjon p.g.a. av ytre aksiallast	[m]
δ_T	defleksjon p.g.a. temperaturforandring	[m]
ΔT	temperaturforandring	[°C ⁻¹]
τ_{max}	maksimal torsjon og skjærspenning	[Pa]
$\tau_{avg.}$	gjennomsnittlig skjærspenning	[Pa]
T	torsjonsmoment	[Nm]
c	avstand fra t.p. til ytterkant i tverrsnittet	[m]
c_i og c_y	indre og ytre radius	[m]
J	polart arealmoment	[m ⁴]
M	bøymoment	[Nm]
w	last pr. meter (kan også være vist som q)	[N/m]
I	annet arealmoment	[m ⁴]
I'	annet arealmoment med Steiners formel	[m ⁴]
d_y og d_x	avstand til tyngdepunkt	[m]
V	skjærkraft	[N]
Q	statisk moment	[m ³]
A'	arealet av flate elementet til tverrsnittet	[m ²]
$\bar{y}'$	avst. fra t.p. i tverrsnitt til t.p i A'	[m]
t	tykkelse	[m]
P_{cr}	maks. last søylen tåler før den knekker	[N]
σ_{cr}	maks. spenning søylen tåler før den knekker	[Pa]
L	lengden på søylen	[m]
K	effektiv lengdefaktor	
r	tregghetsradius (radius of gyration)	[m]
FS	sikkerhetsfaktor (Factor of Safety)	
$S_{req'd}$	nødvendig motstandsmoment (section modulus)	[m ³]
	Pa = N/m ²	
	MPa = N/mm ²	
	GPa = 10 ³ Mpa	
	1 m = 10 ³ mm	
	1 m ² = 10 ⁶ mm ²	
	1 m ³ = 10 ⁹ mm ³	

Dimensjonering:

$$S_{req'd} = \frac{M_{max.}}{\sigma_{allow}} = \frac{I}{c}$$

Skjær:

$$\tau_{avg.} = \frac{V}{A} \quad \tau_{max.} = \frac{V \cdot Q}{I \cdot t} \quad Q = \sum \bar{y} \cdot A'$$

Parallell-akse teoremet (Steiner's formel)

$$I'_x = I_x + A \cdot d_y^2$$

$$I'_y = I_y + A \cdot d_x^2$$

Tyngdepunkt

$$\bar{x} = \frac{x_1 \cdot A_1 + x_2 \cdot A_2 + \dots + x_n \cdot A_n}{A_1 + A_2 + \dots + A_n}$$

$$\bar{y} = \frac{y_1 \cdot A_1 + y_2 \cdot A_2 + \dots + y_n \cdot A_n}{A_1 + A_2 + \dots + A_n}$$

Kombinert spenning

$$\text{Cylinder} \quad \sigma_1 = \frac{pr}{t} \quad \sigma_2 = \frac{pr}{2t} \quad \text{Sphere} \quad \sigma_1 = \sigma_2 = \frac{pr}{2t}$$

Normalspenninger

$$\sigma_s = \sigma_b + \sigma_n \quad \sigma_t = \sigma_b - \sigma_n$$

Spenningstransformasjonsligninger

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta \quad \tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta$$

$$\sigma_{x'} + \sigma_{y'} = \sigma_x + \sigma_y$$

Hovedspenninger

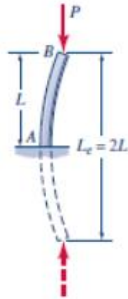
$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \quad \tan 2\theta_p = \frac{\tau_{xy}}{(\sigma_x - \sigma_y)/2}$$

Maks. plan skjærspenning

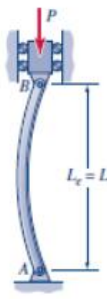
$$\tau_{max} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \quad \tan 2\theta_s = -\frac{(\sigma_x - \sigma_y)/2}{\tau_{xy}}$$

Knekking:

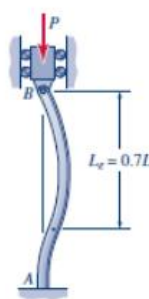
$$P_{cr} = \frac{\pi^2 \cdot E \cdot I}{(K \cdot L)^2} \quad \sigma_{cr} = \frac{\pi^2 \cdot E}{\left(\frac{K \cdot L}{r}\right)^2} \quad r = \sqrt{\frac{I}{A}} \quad FS = \frac{P_{cr}}{P}$$



a) $K = 2$



b) $K = 1$



c) $K = 0,7$



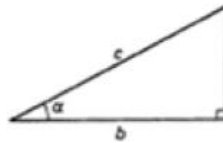
d) $K = 0,5$

- Stav med den ene enden innspent og den andre enden fri.
- Stav med begge ender leddet.
- Stav med den ene enden innspent og den andre enden leddet.
- Stav med begge endene innspent.

TRIGONOMETRI

Pytagoreiske
læresetning

$$c^2 = a^2 + b^2$$



c hypotenus
 a |
 b kateter

Trigonometriske
funksjoner

$$\sin \alpha = \frac{a}{c} \quad \cos \alpha = \frac{b}{c} \quad \tan \alpha = \frac{a}{b}$$

Sinussetningen

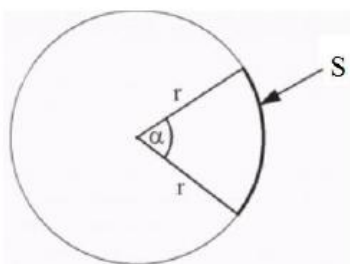
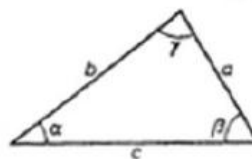
$$\frac{a}{\sin \alpha} = \frac{b}{\sin \beta} = \frac{c}{\sin \gamma}$$

Cosinussetningen

$$a^2 = b^2 + c^2 - 2bc \cos \alpha$$

Tangenssetningen

$$\frac{a-b}{a+b} = \frac{\tan \frac{\alpha-\beta}{2}}{\tan \frac{\alpha+\beta}{2}}$$



$$S = \alpha \cdot r \quad \alpha [\text{rad}]$$

Fundamental Equations of Mechanics of Materials

Axial Load

Normal Stress

$$\sigma = \frac{N}{A}$$

Displacement

$$\delta = \int_0^L \frac{N(x)dx}{A(x)E}$$

$$\delta = \Sigma \frac{NL}{AE}$$

$$\delta_T = \alpha \Delta TL$$

Torsion

Shear stress in circular shaft

$$\tau = \frac{T\rho}{J}$$

where

$$J = \frac{\pi}{2} c^4 \quad \text{solid cross section}$$

$$J = \frac{\pi}{2} (c_o^4 - c_i^4) \quad \text{tubular cross section}$$

Power

$$P = T\omega = 2\pi fT$$

Angle of twist

$$\phi = \int_0^L \frac{T(x)dx}{J(x)G}$$

$$\phi = \Sigma \frac{TL}{JG}$$

Average shear stress in a thin-walled tube

$$\tau_{\text{avg}} = \frac{T}{2tA_m}$$

Shear Flow

$$q = \tau_{\text{avg}} t = \frac{T}{2A_m}$$

Bending

Normal stress

$$\sigma = -\frac{My}{I}$$

Unsymmetric bending

$$\sigma = -\frac{M_z y}{I_z} + \frac{M_y z}{I_y}, \quad \tan \alpha = \frac{I_z}{I_y} \tan \theta$$

Shear

Average direct shear stress

$$\tau_{\text{avg}} = \frac{V}{A}$$

Transverse shear stress

$$\tau = \frac{VQ}{It}$$

Shear flow

$$q = \tau t = \frac{VQ}{I}$$

Stress in Thin-Walled Pressure Vessel

Cylinder

$$\sigma_1 = \frac{pr}{t} \quad \sigma_2 = \frac{pr}{2t}$$

Sphere

$$\sigma_1 = \sigma_2 = \frac{pr}{2t}$$

Stress Transformation Equations

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta$$

$$\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta$$

Principal Stress

$$\tan 2\theta_p = \frac{\tau_{xy}}{(\sigma_x - \sigma_y)/2}$$

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

Maximum in-plane shear stress

$$\tan 2\theta_s = -\frac{(\sigma_x - \sigma_y)/2}{\tau_{xy}}$$

$$\tau_{\text{max}} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma_{\text{avg}} = \frac{\sigma_x + \sigma_y}{2}$$

Absolute maximum shear stress

$$\tau_{\text{abs max}} = \frac{\sigma_{\text{max}}}{2} \quad \text{for } \sigma_{\text{max}}, \sigma_{\text{min}} \text{ same sign}$$

$$\tau_{\text{abs max}} = \frac{\sigma_{\text{max}} - \sigma_{\text{min}}}{2} \quad \text{for } \sigma_{\text{max}}, \sigma_{\text{min}} \text{ opposite signs}$$

Geometric Properties of Area Elements

Material Property Relations

Poisson's ratio

$$\nu = -\frac{\epsilon_{\text{lat}}}{\epsilon_{\text{long}}}$$

Generalized Hooke's Law

$$\epsilon_x = \frac{1}{E} [\sigma_x - \nu(\sigma_y + \sigma_z)]$$

$$\epsilon_y = \frac{1}{E} [\sigma_y - \nu(\sigma_x + \sigma_z)]$$

$$\epsilon_z = \frac{1}{E} [\sigma_z - \nu(\sigma_x + \sigma_y)]$$

$$\gamma_{xy} = \frac{1}{G} \tau_{xy}, \gamma_{yz} = \frac{1}{G} \tau_{yz}, \gamma_{zx} = \frac{1}{G} \tau_{zx}$$

where

$$G = \frac{E}{2(1 + \nu)}$$

Relations Between w , V , M

$$\frac{dV}{dx} = w(x), \quad \frac{dM}{dx} = V$$

Elastic Curve

$$\frac{1}{\rho} = \frac{M}{EI}$$

$$EI \frac{d^4 v}{dx^4} = w(x)$$

$$EI \frac{d^3 v}{dx^3} = V(x)$$

$$EI \frac{d^2 v}{dx^2} = M(x)$$

Buckling

Critical axial load

$$P_{cr} = \frac{\pi^2 EI}{(KL)^2}$$

Critical stress

$$\sigma_{cr} = \frac{\pi^2 E}{(KL/r)^2}, r = \sqrt{I/A}$$

Secant formula

$$\sigma_{max} = \frac{P}{A} \left[1 + \frac{ec}{r^2} \sec \left(\frac{L}{2r} \sqrt{\frac{P}{EA}} \right) \right]$$

Energy Methods

Conservation of energy

$$U_e = U_i$$

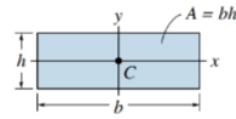
Strain energy

$$U_i = \frac{N^2 L}{2AE} \quad \text{constant axial load}$$

$$U_i = \int_0^L \frac{M^2 dx}{2EI} \quad \text{bending moment}$$

$$U_i = \int_0^L \frac{f_s V^2 dx}{2GA} \quad \text{transverse shear}$$

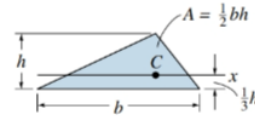
$$U_i = \int_0^L \frac{T^2 dx}{2GJ} \quad \text{torsional moment}$$



Rectangular area

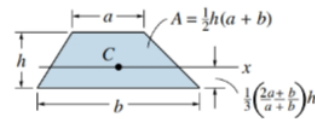
$$I_x = \frac{1}{12} bh^3$$

$$I_y = \frac{1}{12} hb^3$$

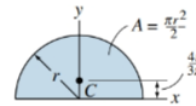


Triangular area

$$I_x = \frac{1}{36} bh^3$$



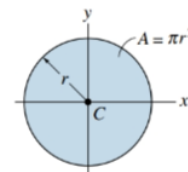
Trapezoidal area



Semicircular area

$$I_x = \frac{1}{8} \pi r^4$$

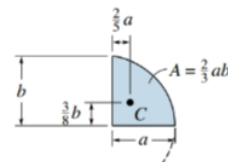
$$I_y = \frac{1}{8} \pi r^4$$



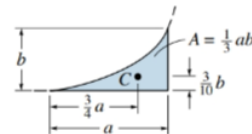
Circular area

$$I_x = \frac{1}{4} \pi r^4$$

$$I_y = \frac{1}{4} \pi r^4$$



Semiparabolic area



Exparabolic area